

# Mohid Tahir

Mechanical Engineer

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## Education

### University of Toronto

Sep. 2021 – Apr. 2026

*Bachelor of Applied Science and Engineering*

*Toronto, ON*

- Program: Mechanical Engineering + PEY Co-op, Minor: Engineering Business, Certificate: EV Design
- Related Coursework: Mechanical Engineering Design, Kinematics and Dynamics of Machines, Electric Vehicle Design, Analog and Digital Electronics, Design for the Environment

## Technical Skills and Qualifications

**Product & Design:** Voice of Customer (VOC), Kano Analysis, House of Quality (HOQ), Competitive Benchmarking, Positioning, Failure Mode Analysis, Iterative Design, Stage-Gate Product Launch, User Research

**Methods & Analysis:** Agile, Six Sigma DMAIC, Lean, Life Cycle Assessment (LCA), NPV Modeling, Weighted Decision Matrices, Cost Modeling, Stakeholder Management

**Software & Tools:** SolidWorks, Microsoft Office Suite, Python, MATLAB, HTML, CSS, Git, AI Tools (Claude, ChatGPT) for rapid prototyping

## Professional Experience

### Design Engineering Intern

May 2023 – Aug. 2023

*Parts Apart*

*Toronto, ON*

- Led end-to-end design of a convertible travel mug (top-mounted carry handle that detaches and reattaches as a side drinking handle) as the sole design engineer, working directly with the founder across discovery, CAD, prototyping, and user testing
- Ran semi-structured Voice of Customer interviews across five user segments (commuter, office, gym, parent, casual) and synthesized findings via affinity mapping, 3/6/9 priority scoring, and Kano categorization
- Benchmarked five competing travel mug formats against eight weighted customer needs and mapped them on a positioning chart, identifying a clear product gap between compactness and handle comfort
- Translated customer priorities into twelve measurable target specifications with tolerances via a House of Quality matrix, and identified four failure modes with mitigation strategies
- Delivered a functional working prototype at under \$50 in material cost, closing the first full design cycle for the product

### Mechanical Engineering Intern (Special Projects)

May 2024 – Aug. 2025

*AGS Automotive Systems*

*Oshawa, ON*

- Modeled 20+ bracket, fixture, and process assemblies in SolidWorks with GD&T-compliant drawings, supporting active production programs across the plant
- Applied the MOST (Maynard Operation Sequence Technique) method to benchmark cycle times on key workflows, identifying changes that cut process time by 15%

### Communications Manager & Project Team Member

Jan. 2022 – Apr. 2022

*Engineering Strategies and Practice II (APS112) - University of Toronto*

*Toronto, ON*

- Served as the single point of contact between a faculty client and a 5-person design team building a teaching device for the course: Six Sigma for Engineers (DMAIC)
- Shortened feedback to revision cycles by roughly 20% through structured weekly touchpoints and a shared requirements log, translating client input into engineering action items

## Projects

### Google Home Housing

Jan. 2024 – Apr. 2024

*Design for the Environment (MIE315)*

*University of Toronto*

- Ran a materials trade study comparing plastic and FSC-certified wood for Google Home speaker housings, using hybrid Life Cycle Assessment and 5-year Net Present Value as the decision basis
- Packaged the analysis into a consulting-style recommendation covering cost, environmental impact, brand perception, and end-of-life disposal

## Awards & Certifications

Dean's Honours List, Certified SolidWorks Associate (CSWA), Basic Machining Certification (George Brown College), University of Toronto Skule Frosh Leader